

Layers

Read the following sentence.

The horse raced past the barn fell.

FIRST READ

It doesn't make sense. Did someone forget a word?

READ IT AGAIN

Wait, did a barn fall? Did the horse race past the barn afterward?

READ IT A THIRD TIME

Got it. A horse ran past a barn. And after running past the barn, the horse fell.

Each pass, your mind updates the meaning until it clicks.

AI DOES THE SAME THING

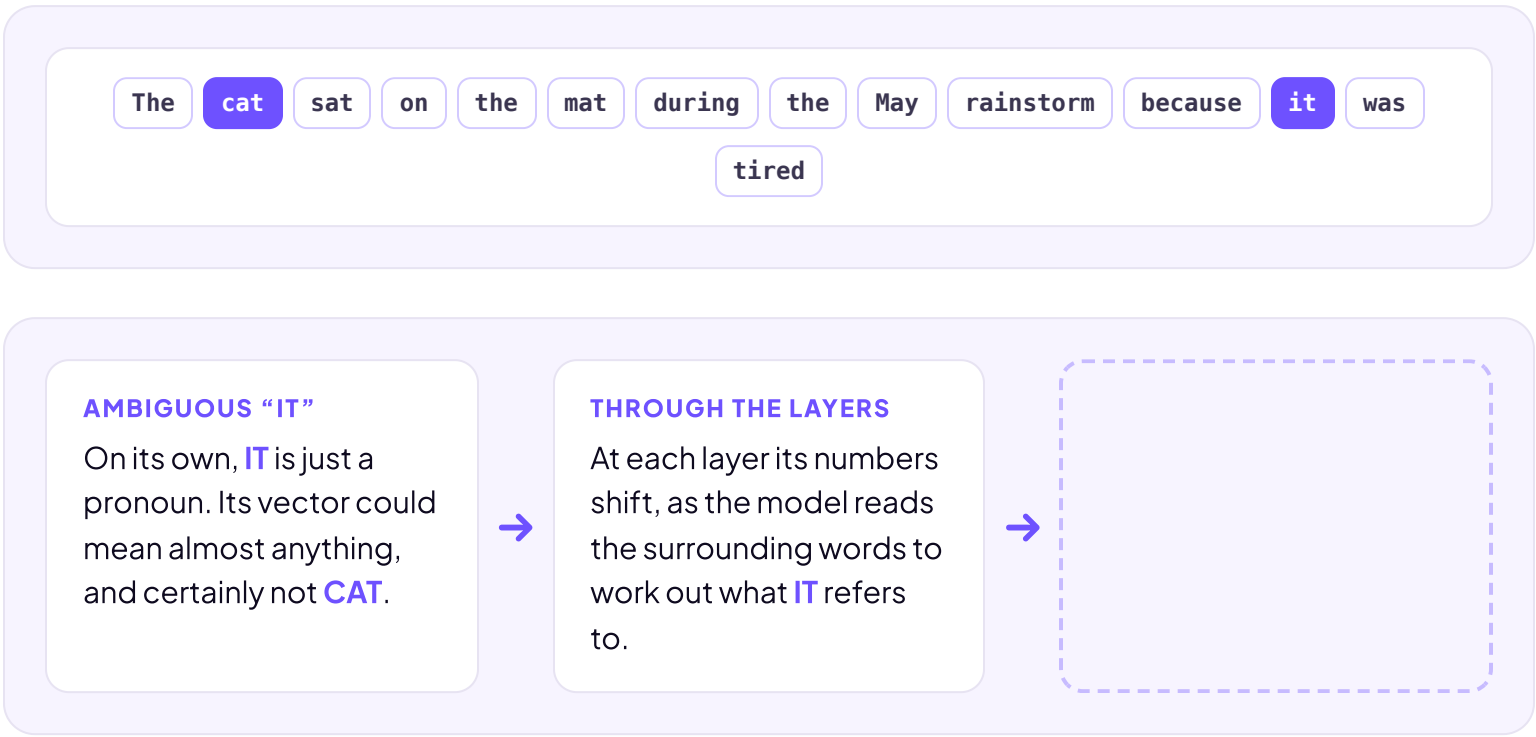
It reads your message over and over, establishing the meaning a little more with each pass. And because language is full of nuance, it takes dozens of passes to pin the meaning down.

Each pass is called a **layer**, and at every layer AI runs the two moves you just met: **attention** (which words matter) and **transformation** (update the meaning).

And AI does it with math. At each layer, it adjusts the tokens' numbers, each pass moving them a little closer to what the tokens mean.

THE MECHANICS

You saw this sentence in the Transformer lesson. Here's how the numbers adjust for one word: **IT**. Notice the last box is blank. That box gets filled in the next lesson.



What does this look like inside the model? Like this.

